

12.744

EE25BTECH11002 - Achat Parth Kalpesh

Question:

Consider the following equations of straight lines:

$$\text{Line L1: } 2x - 3y = 5$$

$$\text{Line L2: } 3x + 2y = 8$$

$$\text{Line L3: } 4x - 6y = 5$$

$$\text{Line L4: } 6x - 9y = 6$$

Which one among the following is the correct statement?

- 1) L1 is parallel to L2 and L1 is perpendicular to L3
- 2) L2 is parallel to L4 and L2 is perpendicular to L1
- 3) L3 is perpendicular to L4 and L3 is parallel to L2
- 4) L4 is perpendicular to L2 and L4 is parallel to L3

Solution:

A line $ax + by = c$ can be written as

$$\mathbf{n}^T \mathbf{x} = c, \quad (4.1)$$

where

$$\mathbf{n} = \begin{pmatrix} a \\ b \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (4.2)$$

Thus, the normal vectors are:

$$\mathbf{n}_1 = \begin{pmatrix} 2 \\ -3 \end{pmatrix}, \quad \mathbf{n}_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \quad \mathbf{n}_3 = \begin{pmatrix} 4 \\ -6 \end{pmatrix}, \quad \mathbf{n}_4 = \begin{pmatrix} 6 \\ -9 \end{pmatrix} \quad (4.3)$$

Two lines are parallel if their normal vectors are proportional:

$$\mathbf{n}_3 = 2\mathbf{n}_1 \implies \text{L1 is parallel to L3,} \quad (4.4)$$

$$\mathbf{n}_4 = \frac{3}{2}\mathbf{n}_3 \implies \text{L3 is parallel to L4.} \quad (4.5)$$

Two lines are perpendicular if

$$\mathbf{n}_i^T \mathbf{n}_j = 0 \quad (4.6)$$

$$\mathbf{n}_1^T \mathbf{n}_2 = \begin{pmatrix} 2 & -3 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 6 - 6 = 0 \implies \text{L1 is perpendicular to L2,} \quad (4.7)$$

$$\mathbf{n}_4^T \mathbf{n}_2 = \begin{pmatrix} 6 & -9 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 18 - 18 = 0 \implies \text{L4 is perpendicular to L2} \quad (4.8)$$

L4 is perpendicular to L2 and L4 is parallel to L3

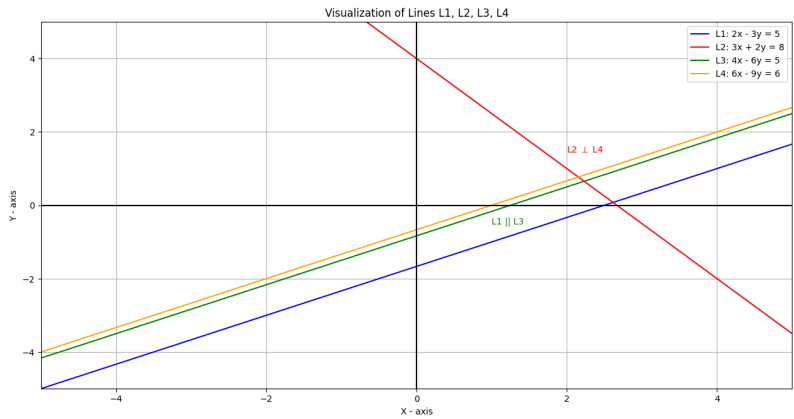


Fig. 4.1: Lines